# 【Proposal】ENGR4903J _Zhongqiang Ren_Mechatronic Design of Snake Robot Modules

**Semester & Year:** 2026 Spring **Instructor:** Zhongqiang Ren

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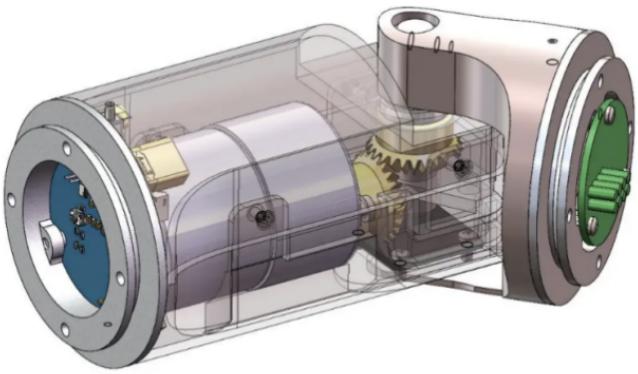
| **Title of Project:**<br>**Mechatronic Design of Snake Robot Modules** |
| **Key Words of the Project:**<br>**Robotics, mechatronic design, embedded systems** |
| **Picture of the Project (for website post):**<br><br>1 DoF revolute joint module |
| **Proposed Credits (1-3 credits):**<br>3 |

**Project Description**:

1. Project Background

Snake robots can access narrow compact space that is otherwise hard to access. Snake robots involve numerous challenges such as motion planning, terrain analysis, mechanical design, etc. This project seeks to develop modules for the snake robot, such as the revolute joint, or the snake head with sensors.

2. Goal of the Research

The goal of the project is to develop modules for the snake robot, such as the revolute joint, or the snake head with sensors, to help the snake robot to navigatae in complex terrains and environments.

3. Primary Approach to Achieve the Goal

Mechanical design, electronic design, and embedded system development.

**Specific Role for the Student:**

The student will be responsible for:

- Learning software and hardware tools related to the project,
- Design and manufacture mechanical parts, circuits, and program the embedded systems,
- Running tests, analyzing the data and
- writing a paper or technical report.

All research activities related to this project will be conducted under the supervision of Prof. Zhongqiang Ren.

The weekly workload of the student is estimated to be around 12 hours.

**Monthly Plan**:

1(st) Month

- Design the parts
- Design the circuits

2(nd) Month

- Prototype the module and program the embedded system

3(rd) Month and onwards

- Design and run experiments, collect test results and analyze the data.
- Write a technical report.

**Student Outcome**:

At the end of the project, the student will:

- Learn knowledge about mechatronic design
- Be able to design and build parts related to robots
- Improve software development and programming skills for embedded systems. Improve writing and presentation skills in academic settings

**Grading Policy** (Include a percentage on each item, weekly progress evaluation and final report must be included):

- Design (25%)
- Prototype (25%)
- Embedded systems (25%)
- Weekly report (10%)
- Final report (15%)

**PREREQUISITES:**

**Major:** All (ECE/ME/MSE) **Class:** All (Sophomore/Junior/Senior)

**Prerequisites (if applicable):**

The student must finish at least one of the following two tasks before being admitted to this project.

https://rap-lab.github.io/documents/prequalifier/mechanical.txt

https://rap-lab.github.io/documents/prequalifier/electronic.txt

**This project belongs to the field of:**

**ECE/ME/MSE/Others (please specify)：Robotics**

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